PHYSICAL CHEMISTRY CUMULATIVE EXAM

October 02, 2015

Chemical equilibria

1. The equilibrium constant for the reaction $N_2(g) + O_2(g) \leftrightarrow 2NO\ (g)$ is K=0.0017 at 2300 K. 5 g N_2 and 2 g O_2 are mixed in a volume of 1 L at 2300 K and allowed to equilibrate.

   (a) What is the mole fraction of NO in the equilibrium gas mixture? [Help: the equilibrium constant is small, and the amounts of N_2 and O_2 in the reaction mixture can be approximated by their initial concentrations which simplifies the algebra. Note that you don't need the gas constant to solve this problem.]

   (b) The standard Gibbs free energy of formation of NO gas at 25 °C is $\Delta_f G^o(NO)$ = +86.55 kJ/mol. Calculate the equilibrium constant of the above reaction at 25 °C. [Help: Do not forget that 2 molecules of NO are formed in the reaction.]

   (c) The standard heat of formation of NO at 25° C is $\Delta_f H^o(NO)$ = +90.25 kJ/mol. Calculate the equilibrium constant of the above reaction at 800° C, assuming that the enthalpy of the reaction does not change in this temperature range. [Help: The van't Hoff equation, describing the temperature dependence of the equilibrium constant, can be written as $d(\ln K)/d\tau = -\Delta_r H^o/R$. where $\tau = 1/T$ is the *inverse absolute temperature*. This means that ln K as a function of τ is a straight line.]

   (d) Which of the following three possibilities is true? Compressing the mixture of N_2 and O_2, i.e., increasing the overall pressure is expected to shift the reaction toward the formation of NO (forward), toward the dissociation of NO (reverse), or have little effect on the quantity of NO formed.

Data: Atomic masses N=14.01 g/mol, O=16.00 g/mol, R=8.3145 J K^{-1} mol^{-1}, 0° C = 273.15 K.